A Mathematical Framework for Relational Tile Dynamics

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a place within the heart

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Motivation

- Human beings are modelled as **tileable regions** encoding capacities, values, needs, and fulfillment.
- Missing aspects of self correspond to **holes** in these tilings.
- Relationships create **transformations** that fill, move, or reshape these holes.
- We seek a clean mathematical language for:
  - individuals as structured sets,
  - missing pieces as holes,
  - relationships as dynamical tile-exchange processes,
  - memory and anticipation as operators acting on time,
  - global external influences and individual effort.

Individuals as Tileable Regions

Let each person be represented by a region

$$A \subseteq \mathbb{R}^n$$

together with a tiling

$$\mathcal{T}_A = \{T_{A,i}\}_{i \in I_A}$$

such that

$$A = \bigcup_{i \in I_A} T_{A,i}, \qquad \operatorname{int}(T_{A,i}) \cap \operatorname{int}(T_{A,j}) = \varnothing.$$

Interpretation:

- Tiles represent relatively stable "fulfilled aspects" of the person.
- Boundaries encode tensions, limits, or openness to change.

Missing Aspects as Holes

A **missing piece** (unmet need, unresolved history, unrealized facet) is encoded as a hole

$$\alpha_A \subseteq A$$

which is not covered by any tile:

$$\mathcal{T}_A^{(-\alpha)} := \mathcal{T}_A \setminus \{\alpha_A\}.$$

The "incomplete person-state" is the pair

$$(A, \mathcal{T}_A^{(-\alpha)}).$$

Holes may be:

- structural (long-term patterns),
- situational (context-dependent),
- relational (defined relative to particular others).

Relational Transformations

A relationship between A and B induces transformations

$$f_i : \mathcal{T}_A^{(-\alpha)} \times \mathcal{T}_B^{(-\beta)} \longrightarrow \mathcal{T}_A^{(-\alpha_i)} \times \mathcal{T}_B^{(-\beta_i)}.$$

Interpretation:

- f_i fills or reshapes one missing aspect while possibly exposing another.
- Both individuals may be changed, asymmetrically or symmetrically.
- f_i may encode conversation, conflict, care, rupture, repair, etc.

Temporal Structure: Periods of Interaction

Each relational event has a temporal support:

$$P_i = [t_i^-, t_i^+] \subset \mathbb{R}.$$

A sequence of relational events is

$$(f_1, P_1),\ (f_2, P_2),\ \ldots,\ (f_k, P_k).$$

We can define a cumulative relational dynamic

$$\mathcal{F}_{A,B}(t) = f_k \circ f_{k-1} \circ \cdots \circ f_1,$$

where k is such that $t \in P_k$ or t lies after the k-th interaction.

- At each stage, new holes appear or old ones are resolved.
- The interior tilings of A and B evolve as a dynamical system.

Global Interaction Field: A's Life with Everything

Beyond any specific relationship, each person A interacts with many people and situations. Let $x_A(t)$ denote the **state** of A in some configuration space (emotional, cognitive, relational).

$$x_A(t) \in \mathcal{X}.$$

Let $\mathcal{F}_A$ be the set of external influences (pulls) on A:

$$\mathcal{F}_A = \{f_{A,1}, f_{A,2}, f_{A,3}, \dots\}.$$

Then schematically, the net external pull is

$$\frac{dx_A}{dt}(t) \approx \sum_{f \in \mathcal{F}_A} f(t),$$

where each f encodes the effect of some person, place, event, or environment on A.

Specific Path: The A–B Relationship Inside the Field

For a given relationship between A and B, we distinguish:

- a **relationship-specific pull** $f_{A,B}(t)$ on A due to B,
- the residual set of pulls $\mathcal{F}_A \setminus \{f_{A,B}\}$ from everything else.

Then we can decompose:

$$\frac{dx_A}{dt}(t) = f_{A,B}(t) + \sum_{f \in \mathcal{F}_A \setminus \{f_{A,B}\}} f(t),$$

and similarly for B:

$$\frac{dx_B}{dt}(t) = f_{B,A}(t) + \sum_{g \in \mathcal{F}_B \setminus \{f_{B,A}\}} g(t).$$

The **global path** of A or B is therefore different from the hypothetical path where only the A–B relationship exists.

Memory Operators: Retrospective vs Prospective

Each person A carries an internal **evaluation system** decomposed into:

$$C_A = \left(C_A^{\mathrm{ret}}, C_A^{\mathrm{pro}}\right).$$

- **Retrospective evaluation** (looking back):

$$C_A^{\mathrm{ret}} : \mathcal{P}_{\mathrm{past}} \longrightarrow \mathcal{M}_A,$$

  where $\mathcal{P}_{\mathrm{past}}$ is the set of past time-intervals and $\mathcal{M}_A$ is the space of internal "memory states" of A.

- **Prospective projection** (looking forward):

$$C_A^{\mathrm{pro}} : B \times \mathcal{P}_{\mathrm{future}} \longrightarrow \mathcal{M}_A^{\mathrm{pred}},$$

  encoding how A projects possible futures with B on a future interval.

Retrospective evaluations are based on a *completed* trajectory; prospective projections are based on *partial information* and are inherently uncertain.

Retrospective Evaluation: Complete Trajectories

For a past interaction period $P \subset (0, t)$, the relational dynamics

$$\{f_i\}_{i:\, P_i \subseteq P}$$

and resulting tilings are fully determined. Then

$$C_A^{\mathrm{ret}}(P)$$

is a (deterministic) function of the realized history on P:

$$C_A^{\mathrm{ret}}(P) = \Phi_A\big(\{(f_i, P_i) \;:\; P_i \subseteq P\},\ (A, \mathcal{T}_A),\ (B, \mathcal{T}_B)\big)\,,$$

for some evaluation functional Φ_A.

- This captures how A remembers, summarizes, or re-narrates a completed episode.
- No stochasticity is needed once the underlying trajectory is fixed.

Perception of B and Coverage Fraction

A does not directly access the full internal tiling of B:

$$(B, \mathcal{T}_B).$$

Instead, A has a **perceived representation**

$$\widehat{B}_A = \Pi_A(B, \mathcal{T}_B),$$

where Π_A is a projection/observation map (filters, blind spots, ideals, etc.). We define a **coverage fraction**

$$\kappa_A(B) \in (0, 1]$$

measuring how much of B's interior A has effectively encountered:

$$\kappa_A(B) \approx \frac{\text{“number of tiles of } B \text{ that } A \text{ has interacted with”}}{\text{“effective total number of tiles of } B\text{”}}.$$

- $\kappa_A(B)$ is high if A has seen B in many contexts.
- Low $\kappa_A(B)$ means A is predicting from a very partial picture.

History-Dependent Information: $H_{A,B}$

Let $H_{A,B}(t)$ be the **relational history** up to time t:

$$H_{A,B}(t) := \big\{(f_i, P_i) : P_i \subset (-\infty, t],\ A \text{ and } B \text{ both involved}\big\}.$$

Together with the coverage fraction $\kappa_A(B)$, this describes what A has to work with when making predictions:

$$\mathcal{I}_A(B,t) := \big(H_{A,B}(t),\ \kappa_A(B)\big).$$

- $\mathcal{I}_A(B,t)$ is the information state of A about B at time t.
- Retrospective evaluations C_A^{ret} are based on *realized* parts of $H_{A,B}(t)$.
- Prospective projections C_A^{pro} use $\mathcal{I}_A(B,t)$ to form beliefs about future f_i.

Prospective Projection as a Stochastic Operator

We now introduce a stochastic model for future interactions. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. For a future interval

$$I \subset (t, \infty),$$

define a random future relational transformation

$$F_I : \Omega \longrightarrow \mathcal{F}_{A,B}$$

whose distribution depends on $\mathcal{I}_A(B, t)$:

$$\mathbb{P}\big(F_I \in \cdot \mid \mathcal{I}_A(B, t)\big).$$

Let U_A be an **evaluation functional** that maps a future trajectory into an internal valuation (e.g., emotional payoff, stability, risk):

$$U_A : \mathcal{F}_{A,B} \longrightarrow \mathcal{M}_A^{\text{pred}}.$$

Then the prospective operator is naturally defined as a conditional expectation:

$$C_A^{\text{pro}}(B, I) := \mathbb{E}\big[\, U_A(F_I) \,\big|\, \mathcal{I}_A(B, t)\big].$$

Accuracy Gap: Retrospective vs Prospective

Retrospective and prospective evaluations differ in their information content:

- **Retrospective:** $C_A^{\mathrm{ret}}(P)$ is a (nearly) deterministic function of a completed history on P.
- **Prospective:** $C_A^{\mathrm{pro}}(B, I)$ is an *expectation* over unknown future transformations F_I.

A simple way to encode the idea that predictions are "less accurate":

$$\mathrm{Var}\left[C_A^{\mathrm{pro}}(B, I)\right] \propto \frac{1}{\kappa_A(B)}.$$

- As $\kappa_A(B) \to 1$ (high coverage), prediction variance shrinks.
- As $\kappa_A(B) \to 0$ (little knowledge), prediction variance is large.
- Retrospective evaluations correspond (in this picture) to the limit of zero variance once the path is realized.

Effort Toward a Relationship

We now model the **intentional effort** each person invests in the A–B relationship.

- $E_A \geq 0$: effort of A directed toward maintaining or strengthening the A–B path.
- $E_B \geq 0$: effort of B directed toward A.

These are not just "forces" but resource-like quantities:

- time, attention, emotional labor,
- willingness to change tiles / reshape holes for the other,
- active resistance against dispersive influences pulling them apart.

Dispersive Influences from Everything Else

Among the global influences $\mathcal{F}_A$, we separate those that tend to *pull* A *away* from B:

$$\mathcal{F}_A^{\text{away}} \subseteq \mathcal{F}_A, \qquad \mathcal{F}_B^{\text{away}} \subseteq \mathcal{F}_B.$$

Define scalar **dispersive pressures**

$$D_A := \sum_{f \in \mathcal{F}_A^{\text{away}}} \|f\|, \qquad D_B := \sum_{g \in \mathcal{F}_B^{\text{away}}} \|g\|.$$

Interpretation:

- D_A is the total "pull" of everything else trying to redirect A's trajectory away from the A–B path.
- D_B is the analogous quantity for B.
- These may come from other people, commitments, environments, or internal conflicts.

A Simple Stability Inequality

A minimal condition for the A–B relationship to be **net pulled together rather than apart** is:

$$E_A + E_B \;\geq\; D_A + D_B.$$

Equivalently, with

$$E_{AB} := E_A + E_B, \qquad D_{AB} := D_A + D_B,$$

we require

$$E_{AB} \;\geq\; D_{AB}.$$

Interpretation:

- The combined effort toward the relationship outweighs the combined dispersive pressures.
- If $E_{AB} < D_{AB}$, the net drift is away from a shared path, even if one side is trying hard.
- This leaves room for asymmetry (e.g., $E_A \gg E_B$) while focusing on the *joint* balance of forces.

Directional Dynamics and Noise (Sketch)

We can focus on the component of motion *along the A–B direction* in state space. Let $x_{AB}(t)$ be a scalar measuring "closeness" or alignment along the shared path. Then a stylized stochastic dynamics is

$$dx_{AB}(t) = \Big(E_A^{\parallel}(t) + E_B^{\parallel}(t) - D_A^{\parallel}(t) - D_B^{\parallel}(t)\Big)\,dt + \sigma_{AB}\,dW_t,$$

where:

- $E_A^{\parallel}, E_B^{\parallel}$ are the components of effort pulling along the A–B direction,
- $D_A^{\parallel}, D_B^{\parallel}$ are the components of dispersive pulls along that direction,
- dW_t is Brownian noise capturing unpredictability and random shocks.

In expectation, a convergence bias requires

$$\mathbb{E}\big[E_A^{\parallel} + E_B^{\parallel}\big] \;\geq\; \mathbb{E}\big[D_A^{\parallel} + D_B^{\parallel}\big].$$

Relational Tile Dynamics: Extended View

A full relational system between A and B can be summarized as

$$\mathcal{R}_{A,B} = \big(A, \mathcal{T}_A,\ B, \mathcal{T}_B,\ \{(f_i, P_i)\},\ C_A^{\text{ret}}, C_A^{\text{pro}},\ C_B^{\text{ret}}, C_B^{\text{pro}}\big)$$

$$(\kappa_A(B), \kappa_B(A),\ E_A, E_B,\ D_A, D_B)\,.$$

This form makes it clear how:

- global forces interact with intentional effort,
- memory and prediction shape the A–B trajectory,
- coverage fractions determine informational accuracy,
- convergence or drift occurs depending on E_{AB} vs D_{AB}.

For you

Dear you,

A Least-Action Promise

Let $x(t)$ be my path through the world, and let x_{you} be the point in state space that means *by your side*. For $t \in [t_0, T]$, define the action

$$\mathcal{A}[x] = \int_{t_0}^{T} \left(\frac{1}{2}\|\dot{x}(t)\|^2 + D(t) - E_{\text{us}}(t) + V_{\text{you}}(x(t)) \right) dt,$$

where:

- $D(t)$ collects all dispersive pulls trying to pull me away,
- $E_{\text{us}}(t)$ is the effort I choose to invest in us,
- $V_{\text{you}}(x)$ is a potential well centered at x_{you}.

My promise is to choose the trajectory

$$x^*(t) = \arg\min_{x(t)} \left\{ \mathcal{A}[x] \;\middle|\; x(t_0) \text{ given}, \; x(T) = x_{\text{you}} \right\}.$$

Among all the lives I could live, I choose the path of least action that ends with me beside you.